



Technical note

Delaunay Hodge star

Anil N. Hirani^{a,*}, Kaushik Kalyanaraman^a, Evan B. VanderZee^b^a Department of Computer Science, University of Illinois at Urbana-Champaign, 201 N. Goodwin Avenue, Urbana, IL, United States^b Argonne National Laboratory, 9700 S Cass Ave, Lemont, IL, United States

ARTICLE INFO

Keywords:

Discrete exterior calculus

Primal mesh

Circumcentric dual

ABSTRACT

We define signed dual volumes at all dimensions for circumcentric dual meshes. We show that for pairwise Delaunay triangulations with mild boundary assumptions these signed dual volumes are positive. This allows the use of such Delaunay meshes for Discrete Exterior Calculus (DEC) because the discrete Hodge star operator can now be correctly defined for such meshes. This operator is crucial for DEC and is a diagonal matrix with the ratio of primal and dual volumes along the diagonal. A correct definition requires that all entries be positive. DEC is a framework for numerically solving differential equations on meshes and for geometry processing tasks and has had considerable impact in computer graphics and scientific computing. Our result allows the use of DEC with a much larger class of meshes than was previously considered possible.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

Discrete Exterior Calculus (DEC) is a framework for numerical solution of partial differential equations on simplicial meshes and for geometry processing tasks [1,2]. DEC has had considerable impact in computer graphics and scientific computing. It is related to finite element exterior calculus and differs from it in how inner products are defined. The main objects in DEC are p -cochains, which for the purpose of this paper may be considered to be a vector of real values with one entry for each p -dimensional simplex in the mesh. For p -cochains a and b their inner product in DEC is $a^T *_{\mathcal{P}} b$ where $*_{\mathcal{P}}$ is a diagonal discrete Hodge star operator. This is a diagonal matrix of order equal to the number of p -simplices and with entries that are ratios of volumes of p -simplices and their $(n - p)$ -dimensional circumcentric dual cells.

For this to define a genuine inner product the entries have to be positive. Simply taking absolute values or considering all volumes to be unsigned does not lead to correct solutions of partial differential equations. See Fig. 1 to see the spectacular failure when unsigned volumes are used in solving Poisson's equation in mixed form. That figure also shows the importance of the Delaunay property and of our boundary assumptions and the success of the signed dual volumes for such meshes that we describe in this paper.

When DEC was invented it was known that completely well-centered meshes were sufficient but perhaps not necessary for defining the Hodge star operator. (A completely well-centered mesh is one in which the circumcenters are contained within the

corresponding simplices at all dimensions. Examples are acute-angled triangle meshes and tetrahedral meshes in which each triangle is acute and each tetrahedron contains its circumcenter.) For such meshes, the volumes of circumcentric dual cells are obviously positive. For some years now there has been numerical evidence that for codimension 1 duality some (but not all) pairwise Delaunay meshes yield positive dual volumes if volumes are given a sign based on some simple rules. Moreover, these seem to yield correct numerical solutions for a simple partial differential equation. A pairwise Delaunay mesh (of dimension n embedded in $\mathbb{R}^N$, $N \geq n$) is one in which each pair of adjacent n -simplices sharing a face of dimension $n - 1$ is Delaunay when embedded in $\mathbb{R}^n$. (Imagine a pair of triangles with a hinge at the shared edge and lay the pair flat on a table.) This generalizes the Delaunay condition to triangle mesh surfaces embedded in three dimensions and analogous higher dimensional settings. For planar triangle meshes and for tetrahedral meshes in three dimensions pairwise Delaunay is same as Delaunay.

We give a sign convention for dual cells and a mild assumption on boundary simplices. With these in hand, for pairwise Delaunay meshes it is easy to see that the codimension 1 dual lengths are positive in the most general case (dimension n mesh embedded in $\mathbb{R}^N$). In addition, we prove that such triangle meshes embedded in two or three dimensions have positive vertex duals and that the duals of vertices and edges of tetrahedral meshes in three dimensions are positive. This settles the question of positivity for all dimensions of duality for simplicial meshes used in physical and graphics applications and opens up the possibility of using DEC with a much larger class of meshes.

Note that the results of this paper are relevant for the *assembly* of the duals from elementary duals (defined in Section 2). Thus it

* Corresponding author.

E-mail address: hirani@illinois.edu (A.N. Hirani).

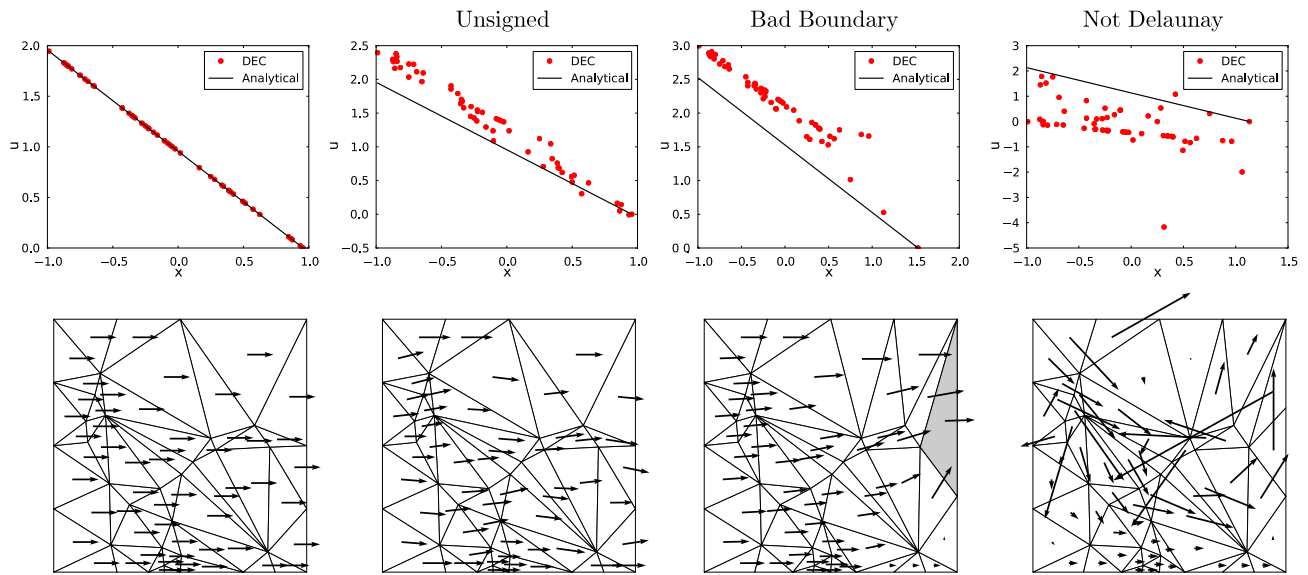


Fig. 1. Solution of Poisson's equation $-\Delta u = f$ in mixed form. In mixed form this equation is the system $\sigma = -\text{grad } u$ and $\text{div } \sigma = f$. The boundary condition is constant influx on left and outflux on right. The correct solution is linear u which varies only along x -direction and a constant horizontal σ . The top row shows u and bottom row shows σ . The first column shows the correct solution using the results of this paper on a Delaunay mesh with correct boundary simplices. The next three columns show various failure modes of alternatives. Second column is for unsigned duals using the same mesh as first column. The third column has a single bad (i.e., not one-sided—see Section 4) boundary triangle shown shaded in a Delaunay mesh. The fourth column is a non-Delaunay mesh.

is important for algorithms which compute these dual volumes piece by piece from the elementary duals (such as are used in the software PyDEC [3]). It may be possible to have alternative formulas which bypass the assembly process and give the dual volumes directly. Such formulas are hinted at in [4]. Zobel [5] has also hinted at the positivity of the duals in some cases but without proofs or details.

There are alternatives to the discrete diagonal Hodge star. In finite element exterior calculus, the role of discrete diagonal Hodge star is played by the mass matrix corresponding to polynomial differential forms [6]. This matrix is in general not diagonal. Sometimes these are referred to as the Galerkin Hodge stars. Our results do not apply to that case directly. For finite elements, there are other quality measures that may be important. See for instance [7].

Notation: We will sometimes write the dimension of a simplex as a superscript. The notation $\tau < \sigma$ means that τ is a face of σ . Circumcenter of a simplex τ is denoted c_τ and the circumcentric dual of τ as $\star\tau$. Some figure labels like $\star p$ stand for dual of a p -simplex.

2. Signed circumcentric dual cells

The dual mesh of a (primal) simplicial mesh is often defined using a barycentric subdivision. For every p -dimensional simplex, there is a dual $(n - p)$ -dimensional cell where n is the dimension of the simplicial complex. In DEC circumcenters are used instead of barycenters because the resulting orthogonality between the primal and dual is an integral part of the definition of some of the operators of DEC [1]. The circumcentric dual cell of a p -dimensional primal simplex τ is constructed from a set of simplices incident to the circumcenter of τ . These are called *elementary dual simplices* with vertices being a sequence of circumcenters of primal simplices incident to τ . The sequence begins with the circumcenter of τ , moves through circumcenters of higher-dimensional simplices σ^i and ends with the circumcenter of a top-dimensional simplex σ^n such that $\tau < \sigma^{p+1} < \dots < \sigma^n$. Taking each of the possibilities for σ^i at each dimension i yields the full dual cell.

In the past in the software PyDEC the volume of the dual cells has been taken to be the sum of unsigned volumes of the elementary dual simplices. Instead we define the volume of the dual cells as the sum of signed volumes of its elementary dual simplices. Our contribution is in defining the sign convention and describing with proofs the class of meshes for which the dual volumes and hence Hodge star entries are positive. If the primal complex is completely well centered every elementary dual has a positive volume. In general, the sign of the volume of an elementary dual simplex is defined as follows. Start from the circumcenter of τ . Let v_p be the vertex such that $v_p \star \tau$ is the simplex σ^{p+1} formed by the vertices of τ together with v_p . Similarly, for $p+1 \leq i \leq n-1$, let v_i be the vertex such that $v_i \star \sigma^i$ is the simplex σ^{i+1} . If the circumcenter of σ^{p+1} is in the same half space of σ^{p+1} as v_p relative to τ , let $s_p = +1$, otherwise, $s_p = -1$. Likewise, for $p+1 \leq i \leq n-1$, if the circumcenter of σ^{i+1} is in the same half space as v_i relative to σ^i , let $s_i = +1$, otherwise, $s_i = -1$. Then, the sign s of the elementary dual simplex is the product of the signs at each dimension, that is, $s = s_p s_{p+1} \dots s_n$.

For illustration of this sign rule, we now consider various cases in two and three dimensions. The first example is the dual of an edge ab in a triangle abc . From the midpoint of ab – its circumcenter – we move to the circumcenter of triangle abc . If this move is towards vertex c , then the sign is $s = s_1 = +1$, but if it is away from vertex c , as it will be if the angle at vertex c is obtuse, then the sign is $s = s_1 = -1$. The next example is the dual of vertex a in triangle abc . We will consider the simplex formed from the circumcenter of a , the circumcenter of ac , and the circumcenter of abc . The move from a to the midpoint of ac gives $s_1 = +1$, since vertex c and the midpoint of ac are in the same direction from a . The move from the midpoint of ac to the circumcenter of abc gives $s_2 = +1$ if we go towards b and $s_2 = -1$ if we move away from b . The sign of the volume of this contribution to the dual of vertex a is $s = s_1 s_2 = s_2$. See Fig. 2 for a graphical illustration.

For a tetrahedron $abcd$ we can expand on the cases for triangle abc . For the dual to face abc , we move from the circumcenter of abc to the circumcenter of $abcd$. If the circumcenter of $abcd$ is in the same half space as vertex d relative to abc , this move is towards d , the sign is $s = s_1 = +1$, and the signed length (volume) is positive;

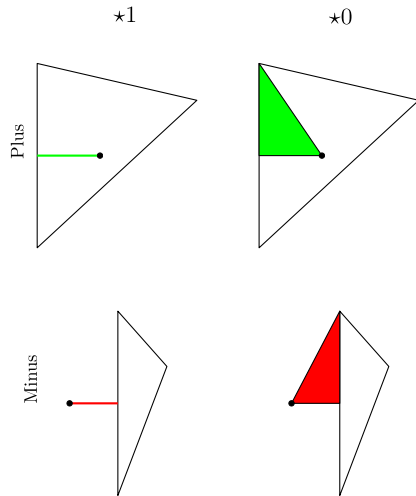


Fig. 2. Examples of sign rule application in 2d. The dot marks the circumcenter and green and red are used to denote positive and negative volumes respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

otherwise, it is negative. Of the two contributions to the dual of edge ab , we focus on the simplex formed from the circumcenter of ab , the circumcenter of abc and the circumcenter of $abcd$. The sign s_1 is determined as it was for the dual of edge ab in triangle abc . The sign s_2 is $+1$ if vertex d and the circumcenter of $abcd$ are in the same half space relative to abc . Thus for the dual of edge ab , the sign of the volume is $s = s_1 s_2$, and both s_1 and s_2 can be either positive or negative. As a final example, consider the simplex formed from vertex a , the circumcenter of ac , the circumcenter of abc , and the circumcenter of $abcd$. This simplex contributes to the dual of vertex a . Signs s_1 and s_2 are the same as they were for the dual of vertex a in triangle abc . Sign s_3 is -1 if triangle abc separates vertex d from the circumcenter of tetrahedron $abcd$. The sign of this elementary volume then is $s = s_1 s_2 s_3$.

The significance of the sign rule defined above is that it orients the elementary dual simplices in a particular way with respect to the dual orientation for a completely well-centered simplex. Consider two n -dimensional simplices σ and σ_w which have the same orientation but such that σ_w is well-centered. We are given a bijection between the vertices of these two simplices such that the resulting simplicial map is orientation preserving. This vertex map induces a bijection between faces of the two simplices and between their elementary duals. Let τ and τ_w be two corresponding p -dimensional faces in the two simplices and consider their duals $\star\tau$ and $\star\tau_w$. If we consider two corresponding elementary duals in $\star\tau$ and $\star\tau_w$ we can affinely map these such that the first vertex (the circumcenter of τ or τ_w) is mapped to the origin and the others are mapped to $+1$ or -1 along a coordinate axis. For the elementary dual in $\star\tau_w$ we always choose $+1$ for all $n - p$ coordinate axes. For $\star\tau$ we choose $+1$ if the sign along that direction of the elementary dual is positive according to the sign rule described above and -1 otherwise. It is clear (and is easy to show using determinants) that the orientation of the corresponding elementary duals will be same if an even number of -1 directions are used for the elementary dual in $\star\tau$ and the orientations will be opposite otherwise. Thus we have shown the following result.

Theorem 1. With σ , σ_w , τ , and τ_w as above, the orientation of $\star\tau$ is same as that of $\star\tau_w$ if an even number of -1 signs appear according to sign rule and is opposite otherwise.

If the orientation of $\star\tau$ is same as $\star\tau_w$ we will assign a positive volume to $\star\tau$ and otherwise a negative volume.

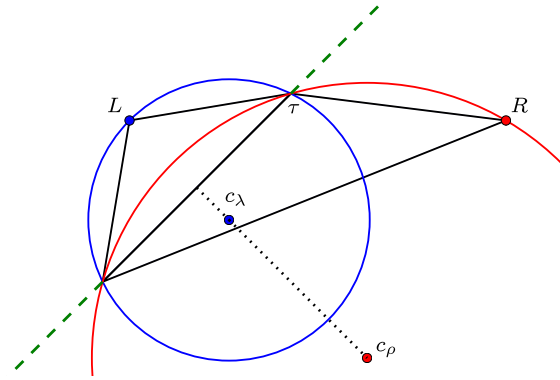


Fig. 3. For a Delaunay pair the ordering of the circumcenters is the same as that of the top dimensional simplices. See Lemma 2.

3. Signed dual of a Delaunay triangulation

We first consider the codimension 1 case in the most general setting of a simplicial complex of arbitrary dimension n embedded in dimension $N \geq n$. After that we consider cases other than codimension 1 but in more restricted settings. For these latter cases we restrict ourselves to the physically most useful cases of triangle meshes embedded in two or three dimensions ($n = 2$ and $N = 2$ or 3) and tetrahedral meshes embedded in three dimensions ($n = N = 3$). We conjecture that these results can be extended to the more general setting of arbitrary n and $N \geq n$ but those cases are not as important for physical applications and we leave those for future work. For the general codimension 1 case we first prove the following basic fact about circumcenter ordering for Delaunay pairs.

Lemma 2 (Circumcenter Order). Let τ be an $(n - 1)$ -dimensional simplex in $\mathbb{R}^n$. Let L and R be points such that $\lambda = L * \tau$ and $\rho = R * \tau$ form a non-degenerate Delaunay pair of n -dimensional simplices with circumcenters c_λ and c_ρ , respectively. Then, c_λ and c_ρ have the same relative ordering with respect to τ as L and R .

Proof. Consider the collection of $(n - 1)$ -dimensional spheres containing the vertices of τ . Since λ and ρ are a non-degenerate Delaunay pair, their circumspheres are empty and belong to this collection. It is then easy to see that c_λ and c_ρ will be in the same order as λ and ρ . See Fig. 3. $\square$

The above lemma can now be used to show easily that the codimension 1 duals always have positive net length. This is the content of the next result.

Theorem 3 (Codimension 1). Let τ be a codimension 1 shared face of two n -dimensional simplices embedded in $\mathbb{R}^N$, $N \geq n$ forming a Delaunay pair. Then the signed length $\star\tau$ is positive.

Proof. When $N = n$, the results directly follows from Lemma 2 since the circumcenters are in the correct order. For $N > n$, we can isometrically embed the simplices in $\mathbb{R}^n$ in which case, the circumcenters are again in the correct order and the result follows. $\square$

In the $N > n$ case, the signs of the elementary dual edges of $\star\tau$ are assigned in the affine spaces of the corresponding n -dimensional simplices. For example, consider a pair of triangles embedded in $\mathbb{R}^3$ and meeting at a shared edge at an angle other than π . In this case, the signed length of the dual edge of the shared edge is determined as the sum of the two elementary dual edges which are measured in the planes of the two triangles individually.

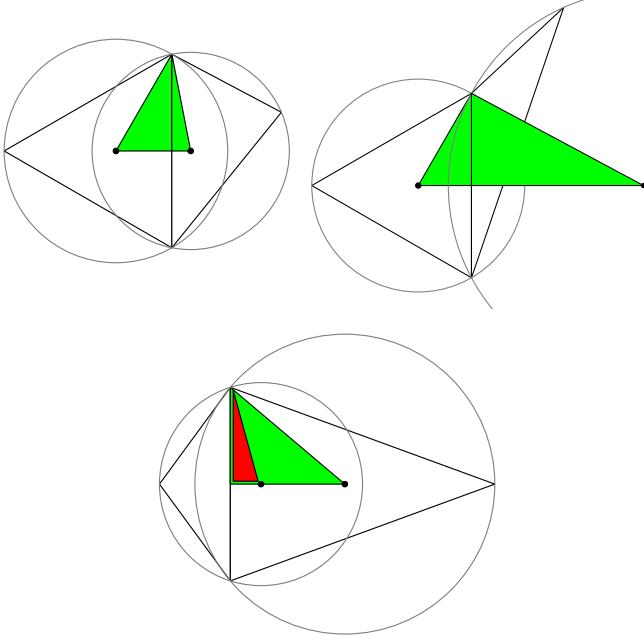


Fig. 4. Elementary dual simplices of a vertex in a pair of triangles sharing an edge. The cases shown correspond to various positions of the circumcenters of the shared edge and the two triangles.

3.1. Dual of a vertex in triangle mesh surface

Now we show that the area of the dual of an internal vertex in a pairwise Delaunay triangle mesh is always positive. We prove this below by showing that the net dual area corresponding to a pair of triangles is positive.

Theorem 4. Let τ be an internal vertex in a pairwise Delaunay triangle mesh embedded in $\mathbb{R}^N$, $N = 2, 3$. Then the signed area of $\star\tau$ is a positive.

Proof. $\star\tau$ is the Voronoi cell of vertex τ in the pairwise Delaunay mesh. Consider a pair of triangles sharing a common edge incident to τ and if they are embedded in $\mathbb{R}^3$, isometrically project to $\mathbb{R}^2$ (i.e., treat the shared edge as a hinge, and flatten the pair). The circumcenters of these two triangles are in correct order by Lemma 2 and there are three possible cases as shown in Fig. 4. Thus the net area of the two elementary dual simplices is positive when the signs are assigned using the rule described in Section 2. Summing over all edges containing τ yields the full $\star\tau$ as a positive area. $\square$

3.2. Dual of an edge in tetrahedral mesh

Theorem 5. Let τ be an internal edge in a tetrahedral Delaunay triangulation embedded in $\mathbb{R}^3$. Then $\star\tau$ is a simple, planar, convex polygon whose signed area is positive.

Proof. $\star\tau$ of an internal edge τ in a Delaunay triangulation may or may not intersect τ . The vertices of $\star\tau$ are circumcenters of tetrahedra incident to τ and the boundary edges of $\star\tau$ are dual edges of triangles incident to τ . Note that $\star\tau$ is the interface between the Voronoi cells corresponding to the two vertices of τ and thus is a bounding face of both Voronoi cells. Since the Voronoi cell of a vertex is a convex polyhedron [8], $\star\tau$ is simple, planar and convex.

Suppose τ intersects $\star\tau$. Then the tetrahedra incident to τ and the edges of $\star\tau$ have to be in a configuration shown in left part

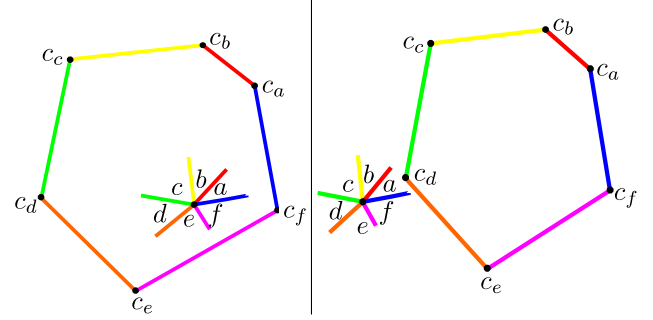


Fig. 5. An internal edge τ of a tetrahedral mesh may or may not intersect $\star\tau$. The views here are along τ which appears as a point. The short lines are half-planes of the triangles incident to τ . The tetrahedra are labeled a, b , etc. Each boundary edge of $\star\tau$ corresponds to the triangle indicated by the coloring. The half planes could potentially be a reflection about τ but that is impossible in a Delaunay mesh due to Lemma 2.

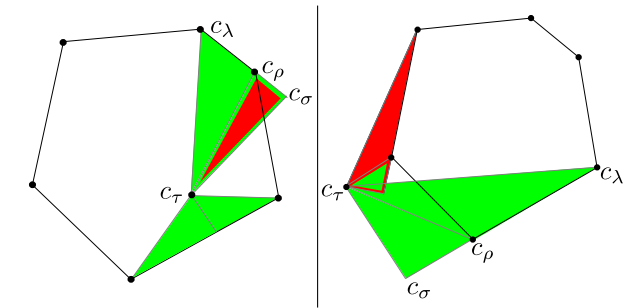


Fig. 6. Representative elementary dual simplices of $\star\tau$ when it intersects τ (left side) and does not intersect τ (right side) corresponding to the two cases shown in Fig. 5.

of Fig. 5. A configuration in which the triangles incident to τ are reflected about τ is impossible due to Lemma 2.

Now, to see that the signed area of $\star\tau$ is positive, consider two elementary dual simplices of $\star\tau$ incident to a shared face σ of two tetrahedra in the fan of tetrahedra incident to τ . These two elementary dual simplices can be in one of the two configurations as shown in Fig. 6. In both cases, c_τ is the circumcenter of the edge τ , c_σ is the circumcenter of the shared face σ , and c_ρ and c_λ are the circumcenters of the two tetrahedra. Also, in both cases, using the sign rule of Section 2 the sum of the signed areas of the elementary dual simplices is positive, and hence, the signed area of $\star\tau$ composed of these elementary dual simplices is positive.

Next consider the case in which τ does not intersect $\star\tau$ as shown in right part of Fig. 5. A boundary edge of $\star\tau$ is called *near side* if it is visible from the midpoint of τ , otherwise, it is called a *far side* edge. Fig. 6 shows the net dual simplices of a near side and far side boundary edge of $\star\tau$. By the sign rule of Section 2, far side elementary dual simplices have a net positive signed area while near side elementary dual simplices have a net negative signed area. The negative areas of the near side dual simplices are covered by the positive areas of the far side dual simplices. Thus, the sum of all these elementary dual simplices which is the signed area of $\star\tau$ is positive. $\square$

3.3. Dual of a vertex in tetrahedral mesh

Theorem 6. Let τ be an internal vertex of a tetrahedral Delaunay mesh embedded in $\mathbb{R}^3$. Then the volume of $\star\tau$ is positive.

Proof. $\star\tau$ of a vertex τ in a Delaunay tetrahedral mesh is a convex polyhedron that is the Voronoi dual cell of τ [8] and thus τ is inside $\star\tau$. The faces of $\star\tau$ are duals of edges incident to τ . By

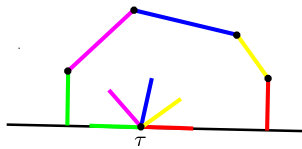


Fig. 7. Dual of an edge τ lying in the boundary of a Delaunay tetrahedral mesh. The meaning of colors and small lines is as in Fig. 5.

Theorem 5 all these faces have a positive signed area. The direction corresponding to traversal from τ to an edge center always has a positive sign. Thus each pyramid formed by τ and a boundary face of $\star\tau$ has positive volume. Thus, the volume of $\star\tau$ is positive. $\square$

4. Requirements on boundary simplices

In the previous section we have only considered internal simplices in a pairwise Delaunay mesh. For simplices lying in the boundary of a domain we require an assumption to ensure positive duals. We call a simplex σ *one-sided* with respect to a codimension 1 face τ if its circumcenter c_σ lies in the same half space as the apex with respect to τ in the affine space of σ .

We show below that the only assumption then needed is that a top dimensional simplex with a codimension 1 face in the domain boundary should be one-sided with respect to the boundary face.

Consider a pairwise Delaunay mesh of dimension n embedded in $\mathbb{R}^N$, $N \geq n$. Assume that τ is an $(n-1)$ -dimensional face appearing in domain boundary and $\tau \prec \sigma^n$ such that σ^n is one-sided with respect to τ .

Theorem 7. For a mesh such as above, duals of codimension one faces have positive lengths. For $n = 2$ and $N = 2$ or 3 , and for $n = N = 3$, duals of all simplices at all dimensions have positive areas or volumes.

Proof. The codimension 1 dual of τ in all cases has positive length using our sign rule since σ^n is one-sided with respect to τ . As a result, for a surface triangle mesh, that is $n = 2$ and $N = 2$ or 3 , it easily follows from our sign rule in Section 2 that the dual of a vertex on the boundary also has a positive area.

For $n = N = 3$, one configuration for the dual of an edge τ incident to the boundary is shown in Fig. 7. In this figure, the plane containing the codimension 1 faces incident to τ are shown as short line segments, and the coloring of boundary edges of $\star\tau$ show the corresponding codimension 1 face they are dual to. The other configuration in which the planes containing the faces incident to τ are mirror images of ones shown is not possible since then the circumcenters of tetrahedra will not be in the correct order as in Lemma 2. Thus, by our sign rule, all elementary dual simplices of $\star\tau$ are positive and hence the signed area of $\star\tau$ is positive. Finally, it follows from our sign rule that the dual of a vertex on the boundary is also positive since each of the elementary dual pyramids will have a positive volume. $\square$

5. Conclusions and outlook

For planar triangle meshes and for tetrahedral meshes in three dimensional space the condition of being pairwise Delaunay is equivalent to being Delaunay. Thus most commercial and freely available meshing software can generate such meshes. In

our experience, several codes for planar meshing also generate meshes for which the one-sidedness condition on the boundary is satisfied. For example, the popular meshing code called Triangle has an option for conforming Delaunay triangulation which generates Delaunay meshes with one-sided boundary triangles. For tetrahedral meshes with acute input angles this property may be harder to achieve. In general however, algorithms for creating tetrahedral meshes with one-sided boundary tetrahedra do exist [9–11]. Note that one-sidedness is equivalent to an “oriented” Gabriel property (using diametral half-balls) for the boundary faces.

The pairwise Delaunay condition also appears to be more natural for DEC than other conditions that are used in place of Delaunay in the case of surfaces. For example, some researchers require that the equatorial balls of triangles not contain another vertex. This disqualifies surfaces with many folds or sharp turns. Another alternative is to define intrinsic Delaunay condition based on geodesics on the triangle mesh but algorithms for such surfaces can be complicated to implement. Yet another alternative is to use Hodge-optimized triangulations [4]. But creation of these requires an additional optimization step. On the other hand Hodge-optimized triangulation is a very interesting generalization of Voronoi–Delaunay duality with many applications.

The invention of algorithms that generate pairwise Delaunay surface meshes is left for future work. So is the proof of our conjecture that the case of codimension other than 1 has positive volume for general dimension and embedding space for pairwise Delaunay meshes with one-sided boundary simplices. Nevertheless, the practically important cases have all been settled by this paper.

Acknowledgments

ANH and KK were supported by NSF Grant DMS-0645604. We thank the anonymous referees for pointing out some important references and for their other suggestions.

References

- [1] Hirani AN. Discrete Exterior Calculus, Ph.D. thesis, California Institute of Technology, May 2003.
- [2] Desbrun M, Kanso E, Tong Y. Discrete differential forms for computational modeling. Discrete differential geometry, vol. 38. Birkhäuser Basel; 2008. 287–324.
- [3] Bell N, Hirani AN. PyDEC: algorithms and software for discretization of exterior calculus. ACM Trans. Math. Softw., 2012;39 (1) [in press].
- [4] Mullen P, Memari P, de Goes F, Desbrun M. HOT: Hodge-optimized triangulations. ACM Trans Graph 2011;30:103:1–103:12.
- [5] Zobel V. Spectral analysis of the Hodge Laplacian on discrete manifolds, Master's thesis, Technische Universität Berlin, 2010.
- [6] Arnold DN, Falk RS, Winther R. Finite element exterior calculus: from Hodge theory to numerical stability. Bull Amer Math Soc (NS) 2010;47(2):281–354.
- [7] Shewchuck JR. What is a good linear element? Interpolation, conditioning, and quality measures. In: Eleventh International Meshing Roundtable, 2002.
- [8] Edelsbrunner H. Geometry and topology for mesh generation. Cambridge University Press; 2006.
- [9] Edelsbrunner H. Surface reconstruction by wrapping finite sets in space. In: Discrete and computational geometry. Algorithms Combin., Springer; 2003.
- [10] Chaîne R. A geometric convection approach of 3-d reconstruction. Proc Sympos Geometry Processing 2003.
- [11] Giesen J, John M. The flow complex: a data structure for geometric modeling. Comput Geom 2008;39(3):178–90.